EDUCATION / PHYSICAL COMPUTING

ESP32 Education Editor

User Guide

Editor v0.6 | Firmware v0.1.7

Target board: ESP32-WROOM-32 / ESP32 Dev Module
Project: davinichi / ESP32 Education Editor
English Edition

ESP32 Education Editor User Guide

> This guide explains the basic setup and use of ESP32 Education Editor for educational physical-computing activities.

1. What is ESP32 Education Editor?

ESP32 Education Editor is an educational block-programming environment for controlling an ESP32-WROOM-32 from a web browser. Firmware v0.1.7 runs on the ESP32, while the editor sends commands over USB / Web Serial.

Editor v0.6 **Firmware v0.1.7** **ESP32-WROOM-32**

- Public Editor: https://davinichi.github.io/
- Firmware Installer: https://davinichi.github.io/firmware/

System Overview



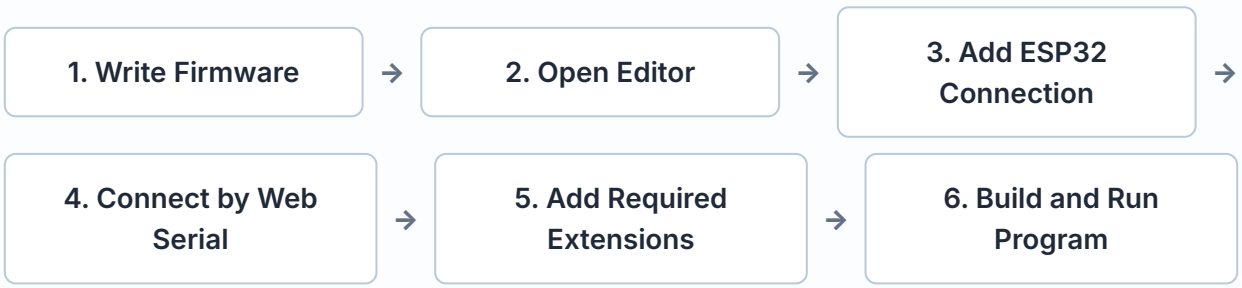
2. What You Need

- ESP32-WROOM-32
- A USB cable that supports data communication
- A Web Serial compatible browser such as Chrome or Edge
- Internet access when loading the public Editor or Firmware Installer
- Sensors, LEDs, OLEDs, servo motors, or other components required by the activity
- Appropriate resistors and power supplies for the circuit

> **Voltage warning:** The HC-SR04 ECHO output is approximately 5 V. Do not connect it directly to an ESP32 input. Reduce the signal to 3.3 V or below.

3. First-Time Setup

Basic Workflow



3.1 Write the Firmware

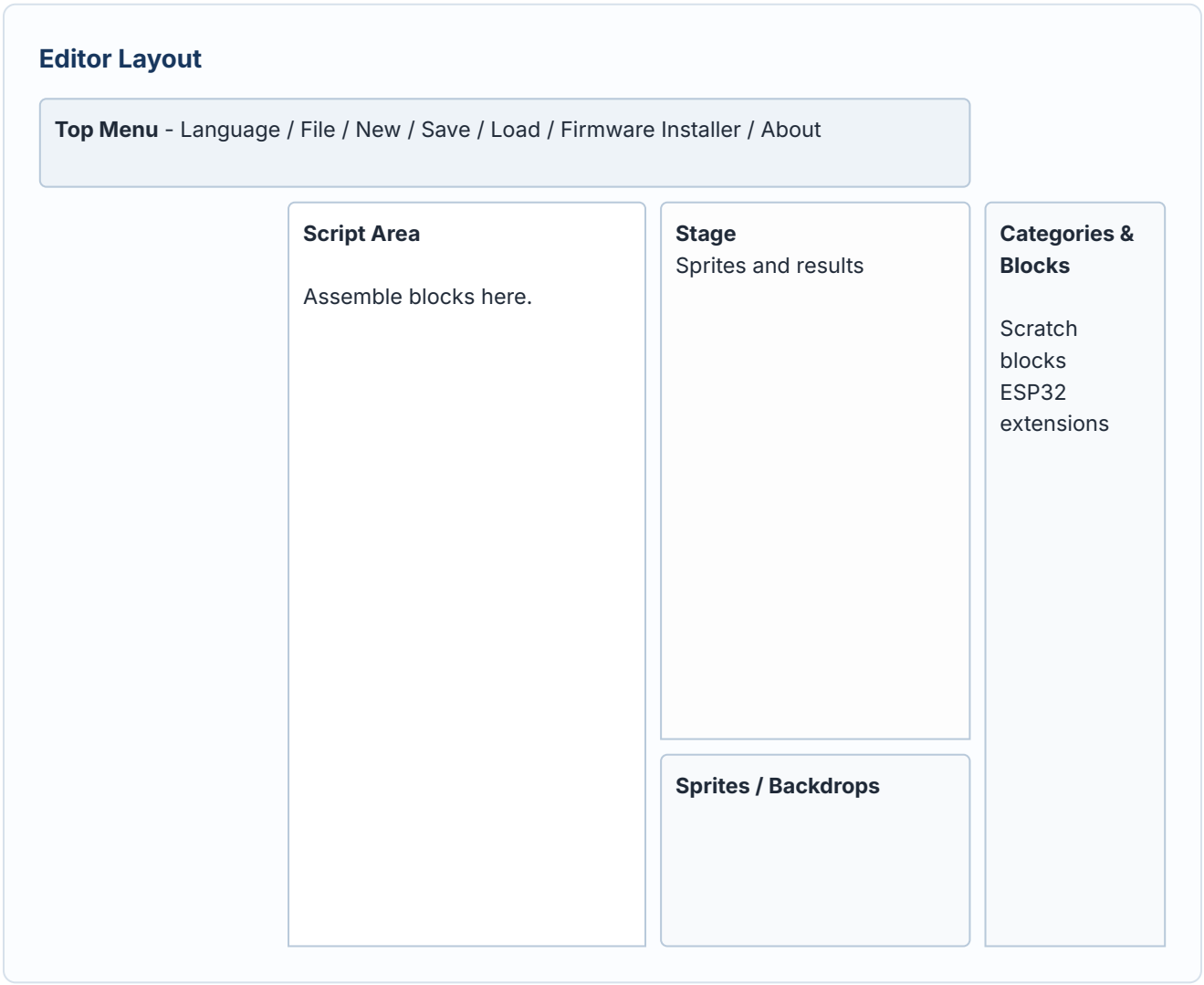
1. Connect the ESP32-WROOM-32 to the PC by USB.
2. Open the Firmware Installer at https://davinichi.github.io/firmware/.
3. Choose the option to connect to the ESP32 and write Firmware v0.1.7.
4. Select the ESP32 USB serial port.
5. Follow the on-screen instructions to complete the installation.

You do not need to prepare Arduino IDE or additional Arduino libraries on the user PC.

3.2 Open the Editor

Open https://davinichi.github.io/ in Chrome or Edge. Use the language menu at the top of the editor to switch between Japanese and English.

4. Editor Layout



- **Top menu:** language, file operations, new project, save/load, Firmware Installer, and About
- **Categories / blocks:** standard Scratch categories plus the added ESP32 extensions
- **Script area:** where blocks are assembled into programs
- **Stage:** displays sprites and program results
- **Sprites / backdrops:** add and select sprites or backgrounds

5. Connect to the ESP32

1. Click the Extensions button at the lower-left of the editor.
2. Add **ESP32 Connection**.
3. Run the **Connect to ESP32** block.
4. Select the ESP32 serial port from the browser chooser.
5. If needed, run **Refresh ESP32 status** and check the status, MAC address, and channel.

If Arduino IDE Serial Monitor or another application is using the same serial port, disconnect or close it before connecting from the browser.

6. Add Extension Categories

Category	Main Use	ESP32 Connection
ESP32 Connection	Connection, status, MAC address, channel	Required
ESP32 GPIO	Digital I/O and PWM	Required
ESP32 DHT	DHT11 / DHT22 temperature and humidity	Required
ESP32 OLED	SSD1306 OLED display	Required
ESP32 ESP-NOW	Text communication between ESP32 devices	Required
ESP32 Servo	Servo motor control	Required
ESP32 Ultrasonic	HC-SR04 distance measurement	Required
ESP32 Environment Index	10 environmental-index calculations	Not required
Data Processing	CSV and text processing	Not required

See the **Block Reference** for details of each block.

7. Building Basic Programs

7.1 LED Example

1. Add ESP32 Connection and connect to the ESP32.
2. Add ESP32 GPIO.
3. Run **set GPIO 23 to output**.
4. Run **write HIGH to GPIO 23** to turn the LED on.
5. Run **write LOW to GPIO 23** to turn the LED off.

Use an appropriate current-limiting resistor with an LED and verify the wiring before powering the circuit.

7.2 Sensor Value Display Example

1. Initialize the DHT sensor.
2. Read DHT temperature and humidity.

3. Initialize the OLED.
4. Set the cursor position and text size.
5. Convert the measured values to text and display them on the OLED.

8. Running Programs

Combine standard Scratch event and control blocks with the ESP32 extension blocks. Typical blocks include:

- when green flag clicked
- forever
- wait ( ) seconds
- if ( ) then
- repeat ( )
- variables and operators

> The current editor uses a live-control model. The PC, browser, and USB connection are required while the program is running.

9. Saving and Loading Projects

Use the **File** menu at the top of the editor.

- **New:** start a new project
- **Load from your computer:** open a saved project
- **Save to your computer:** save the current project locally

The Japanese and English interfaces share the same internal extension IDs and opcodes, so the same project is designed to open in either language.

10. Using ESP-NOW

ESP-NOW allows ESP32 devices to communicate directly.

- School Wi-Fi access point: not required
- School LAN: not required
- Internet during ESP-NOW communication: not required
- ESP32 channel: set the sender and receiver to the same value
- Blank destination MAC: broadcast
- Specified destination MAC: unicast

Internet access is still needed when the public Editor or Firmware Installer is loaded from the web.

11. Hardware Safety and Usage Notes

- The supported target is ESP32-WROOM-32.
- Check the board specification for GPIO numbering, voltage limits, and input-only pins.
- Use a current-limiting resistor with LEDs.
- Check the power requirements of the servo motor used.
- Reduce the HC-SR04 ECHO signal to 3.3 V or below.
- Disconnect power or USB when changing wiring when appropriate.
- Avoid assigning the same GPIO to multiple functions at the same time.

12. Troubleshooting Checklist

1. Is the board an ESP32-WROOM-32?
2. Has Firmware v0.1.7 been written to the board?
3. Are you using Chrome or Edge?
4. Does the USB cable support data communication?
5. Is another application using the serial port?
6. Did you add the extension required by the program?
7. Are the sensor type, GPIO numbers, wiring, and power correct?

13. Related Documents

- ESP32 Education Editor Block Reference
- Getting Started (project document)
- FAQ / Troubleshooting (project document)
- Firmware Guide (project document)
- Lesson / Educational Use Guide (project document)
- Backup / Restore Guide (project document)

At the time of this English edition, some related project documents may still be available only in Japanese.

14. Project Positioning

ESP32 Education Editor is an independent project based on the open-source Scratch Editor published by the Scratch Foundation. It is not an official product of the Scratch Foundation and does not imply endorsement, recommendation, or sponsorship by the Scratch Foundation.